



Shree Shankar Narayan Education Trust's
PRAVIN ROHIDAS PATIL COLLEGE OF
ENGINEERING AND TECHNOLOGY

(Affiliated to the University of Mumbai, Recognised by DTE Govt. of Maharashtra & Approved by AICTE, New Delhi)

DTE CODE - 3277

Experiment No : 5

Aim: -Write a code for game playing algorithms i) Tic tac toe ii) 8-Puzzle problem

1) Tic tac toe

```
# Tuple to store winning positions.
win_positions = (
(0, 1, 2), (3, 4, 5), (6, 7, 8),
(0, 3, 6), (1, 4, 7), (2, 5, 8),
(0, 4, 8), (2, 4, 6)
)
def game(player):
# display current mesh
print("\n", " | ".join(mesh[:3]))
print("----+----+---")
print("", " | ".join(mesh[3:6]))
print("----+----+---")
print("", " | ".join(mesh[6:]))
# Loop until player valid input cell number.
while True:
try:
ch = int(input(f'Enter player {player}'s choice : "))
if str(ch) not in mesh:
raise ValueError
mesh[ch-1] = player
break
except ValueError:
print("Invalid position number.")
# Return winning positions if player wins, else None.
for wp in win_positions:
if all(mesh[pos] == player for pos in wp):
return wp
```



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```
return None
player1 = "X"
player2 = "O"
player = player1
mesh = list("123456789")
for i in range(9):
    won = game(player)
    if won:
        print("\n", " | ".join(mesh[:3]))
        print("---+---+---")
        print("", " | ".join(mesh[3:6]))
        print("---+---+---")
        print("", " | ".join(mesh[6:]))
        print(f"*** Player {player} won! ***")
        break
    player = player1 if player == player2 else player2
else:
    # 9 moves without a win is a draw.
    print("Game ends in a draw.")
```

OUTPUT:-

```
1 | 2 | 3
---+---+---
4 | 5 | 6
---+---+---
7 | 8 | 9
Enter player X's choice : 5
1 | 2 | 3
---+---+---
4 | X | 6
---+---+---
7 | 8 | 9
```



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Enter player O's choice : 3

1 | 2 | O

---+---+---

4 | X | 6

---+---+---

7 | 8 | 9

Enter player X's choice : 1

X | 2 | O

---+---+---

4 | X | 6

---+---+---

7 | 8 | 9

Enter player O's choice : 6

X | 2 | O

---+---+---

4 | X | O

---+---+---

7 | 8 | 9

Enter player X's choice : 9

X | 2 | O

---+---+---

4 | X | O

---+---+---

7 | 8 | X

*** Player X won! ***



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2) 8-Puzzle problem

```
from collections import deque
def bfs(start_state):
    target = [1, 2, 3, 4, 5, 6, 7, 8, 0]
    dq = deque([start_state])
    visited = {tuple(start_state): None}
    while dq:
        state = dq.popleft()
        if state == target:
            path = []
            while state:
                path.append(state)
                state = visited[tuple(state)]
            return path[::-1]
        zero = state.index(0)
        row, col = divmod(zero, 3)
        for move in (-3, 3, -1, 1):
            new_row, new_col = divmod(zero + move, 3)
            if 0 <= new_row < 3 and 0 <= new_col < 3 and abs(row - new_row) + abs(col - new_col) == 1:
                neighbor = state[:]
                neighbor[zero], neighbor[zero + move] = neighbor[zero + move], neighbor[zero]
                if tuple(neighbor) not in visited:
                    visited[tuple(neighbor)] = state
                    dq.append(neighbor)
    def printSolution(path):
        for state in path:
            print("\n".join(' '.join(map(str, state[i:i+3])) for i in range(0, 9, 3)), end="\n-----\n")
    # Example Usage
    startState = [1, 3, 0, 6, 8, 4, 7, 5, 2]
    solution = bfs(startState)
    if solution:
        printSolution(solution)
        print(f"Solved in {len(solution) - 1} moves.")
    else:
        print("No solution found.")
```



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OUTPUT:

1 3 0

6 8 4

7 5 2

1 3 4

6 8 0

7 5 2

1 3 4

6 8 2

7 5 0

1 3 4

6 8 2

7 0 5

1 3 4

6 0 2

7 8 5

1 3 4

0 6 2

7 8 5

1 3 4

7 6 2

0 8 5

1 3 4

7 6 2

8 0 5



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1 3 4

7 0 2

8 6 5

1 3 4

7 2 0

8 6 5

1 3 0

7 2 4

8 6 5

1 0 3

7 2 4

8 6 5

1 2 3

7 0 4

8 6 5

1 2 3

7 4 0

8 6 5

1 2 3

7 4 5

8 6 0

1 2 3

7 4 5

8 0 6

1 2 3

7 4 5

0 8 6



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1 2 3

0 4 5

7 8 6

1 2 3

4 0 5

7 8 6

1 2 3

4 5 0

7 8 6

1 2 3

4 5 6

7 8 0

Solved in 20 moves.